

When can steady glacier flow overturn isochrones?

A cumulative reverse-shear criterion for full-Stokes flow

Bradley Paul Lipovsky

Department of Earth and Space Sciences, University of Washington, Seattle, Washington, USA

Correspondence: bpl7@uw.edu

Abstract

Ancient ice recovered by the Center for Oldest Ice Exploration from the Allan Hills extends into the Miocene, but steep layers, age reversals and small-scale folds complicate interpretation. Existing steady-flow theory shows that isochrones cannot overturn when horizontal speed increases upward. Full-Stokes flow over rough beds can nevertheless contain local extrusion, making that pointwise condition stronger than necessary. We derive a physical-coordinate signed order factor,

$$\mathcal{J} = \frac{u}{u_{s0}} \left[1 + \dot{a}_0 u_{s0} \int_{x_0}^x \frac{u_z}{u^3} d\xi \right] = -\dot{a}_0 \tau_z.$$

Where layers remain ordered, $\mathcal{J}$ is the reciprocal of the established vertical thinning function. Positive, zero and negative $\mathcal{J}$ denote ordered, vertical and overturned stratigraphy. Splitting vertical shear into normal and reverse parts gives the necessary-and-sufficient condition $\mathcal{R} < 1 + \mathcal{P}$: reverse shear must erase both depositional ordering and accumulated surface-fast shear. Full-Stokes mechanics gives no universal sign guarantee, although shallow-ice flow preserves order. We derive a full-Stokes residual threshold, sufficient bounds for non-stagnant localized obstacles, a first-order cancellation result for periodic roughness about plug flow and a trajectory diagnostic. The advance is an explicit cumulative shear budget and its mechanical consequences, rather than a new state variable or universal no-folding theorem.

Keywords: ice stratigraphy; recumbent folds; Allan Hills; COLDEX; full-Stokes flow; extrusion flow; age equation; blue ice

1 Introduction

The oldest ice can be the least stratigraphically simple. Antarctic blue-ice areas expose ancient glacier ice by combining surface mass loss with upward ice motion, thereby providing access to climatic intervals far older than those recovered from conventional vertical cores (Bintanja, 1999; Spaulding and others, 2012). At the Allan Hills, shallow cores first yielded a stratigraphically

25 disturbed section near 1 Ma (Higgins and others, 2015), then discontinuous atmospheric snapshots
older than 2 Ma (Yan and others, 2019), and most recently Pliocene and Miocene ice and air with
reported ages approaching 6 Ma (Shackleton and others, 2025). These discoveries are central to
the Center for Oldest Ice Exploration (COLDEX) effort to reconstruct atmospheric composition
and Antarctic climate beyond the continuous 800 ka ice-core record.

30 The scientific opportunity is inseparable from a structural problem. Allan Hills ice is strongly
thinned and locally out of order. The ALHIC1901 age–depth relation includes multiple reversals,
while the deepest ice can represent very large time intervals per unit core length (Shackleton
and others, 2025). Three-dimensional electrical conductivity measurements resolve mean layer
dips of approximately 29° and 69° in two recent Allan Hills cores (Kirkpatrick and others,
35 2025). Microstructural observations from ALHIC1901 indicate pervasive simple shear, strongly
elongated bubbles, crystal-orientation stripes, and centimetre-scale kink and z-folds (Stoll and
others, 2026). Radar and flow modelling independently show that the duration and continuity
of the archive vary sharply across the blue-ice area and with basal geometry (Kehrl and others,
2018). Deformation is therefore not a secondary correction to the paleoclimate record; it controls
40 which ages are present, how they are ordered, and how much time a sample averages.

A basic unresolved question is whether a fixed bed obstacle in an otherwise steady glacier
can create the finite-amplitude disturbance required for a recumbent fold. Flow over topography
certainly bends internal layers. Full mechanical calculations predict draping over long-wavelength
topography and overriding of shorter-wavelength relief (Hindmarsh, Leysinger Vieli, Raymond,
45 and others, 2006). Rough-bed Stokes solutions can also contain local extrusion, meaning
that horizontal speed increases downward over part of the ice column (Gudmundsson, 1997a;
Waddington, 2010). At larger bed slopes, separated secondary circulation is possible even in
creeping flow (Gudmundsson, 1997b). These results appear, at first sight, to undermine the
common intuition that a steady obstacle can produce only a standing layer disturbance rather
50 than a recumbent fold.

The apparent contradiction reflects three distinct concepts. A fixed obstacle can generate
curvature; local extrusion can rotate a material tangent in the sense favorable for overturning;
and yet an isochrone can remain globally ordered. Recumbent folding requires a stronger event:
particles deposited in an ordered sequence must reverse their horizontal order along an equal-age
55 surface. Curvature, tangent rotation and order reversal are not equivalent.

The local theory of Waddington and others (2001), hereafter WBA, formalized how a
pre-existing wrinkle is flattened by longitudinal extension or overturned by shear. WBA also
stated the key qualitative principle that layers spending their entire history in a steady-state
flow field are not passively folded; a transient process must first create a disturbance relative
60 to the steady stratigraphy. Jacobson and Waddington (2004) replaced the local criterion with
finite-strain integration along trajectories, and Jacobson and Waddington (2005) showed that
divide migration can place an inherited Raymond arch in a shear regime that turns it into a
recumbent fold. For steady stratigraphy, Parrenin, Hindmarsh, and Rémy (2006) introduced

logarithmic flux coordinates for analytical age solutions, and Parrenin and Hindmarsh (2007) generalized the construction to horizontally non-uniform velocity profiles. Their Appendix A showed that an isochrone cannot pass through vertical when horizontal velocity increases monotonically upward. The same characteristic framework subsequently underpinned numerical age solvers (Hindmarsh, Leysinger Vieli, and Parrenin, 2009; Parrenin, Chung, and others, 2025) and an explicit vertical thinning function (Parrenin, 2013; Parrenin, Chung, and others, 2025).

The monotonic-shear condition is powerful but unnecessarily restrictive. Full-Stokes glacier flow does not impose $u_z \geq 0$ pointwise. The correct question is therefore cumulative: how much reverse shear can a particle experience before the depositional ordering of the isochrone is lost? This paper answers that question for steady two-dimensional flow. We:

- (i) derive a physical-coordinate representation of the signed inverse vertical-thinning factor and its interpretation as horizontal order along an isochrone;
- (ii) express its sign as a cumulative competition between surface-fast and reverse vertical shear;
- (iii) decompose the reverse-shear budget using the full-Stokes stress balance;
- (iv) identify regimes in which ordinary glacier mechanics guarantees order and regimes in which it does not;
- (v) derive a sufficient no-overturning bound for localized obstacles and a first-order cancellation result for periodic small-amplitude roughness; and
- (vi) formulate a direct trajectory diagnostic for full-Stokes models of Allan Hills and other old-ice sites.

The novelty is deliberately narrow. The monotonic-shear no-overturning result is due to Parrenin and Hindmarsh (2007); the order factor derived below is the reciprocal of the vertical thinning function developed by Parrenin (2013) and implemented by Parrenin, Chung, and others (2025); the local wrinkle criterion and finite-strain interpretation are due to WBA and Jacobson and Waddington (2004); and rough-bed extrusion is established (Gudmundsson, 1997a). The contribution here is to expose the sign of the inverse thinning factor as an exact positive–negative u_z/u^3 shear budget and to use that budget to obtain a full-Stokes residual threshold, localized-obstacle bounds, a restricted periodic-cancellation result and a direct trajectory diagnostic. These are reformulations and extensions of established steady-isochrone theory, not claims of a new universal state variable or no-folding theorem.

2 Previous descriptions of stratigraphic folding

2.1 Passive and active folding

Glacier stratification is a material structure produced at the surface and subsequently deformed by flow. Structural glaciology distinguishes passive deformation, in which layers are markers that do not alter the velocity field, from active folding, in which rheological or anisotropic contrasts feed back on stress and motion (Hudleston, 2015; Jennings and Hambrey, 2021).

The distinction matters because a prescribed smooth velocity field can amplify or overturn an existing perturbation without selecting its wavelength, whereas active deformation can generate disturbances and select spatial scales.

Microstructural folds in deep ice provide evidence for the active route. At GISP2, crystal-orientation stripes are associated with localized shear and grain-scale folding, and spatial variation in stripe density may create larger viscosity contrasts (Alley and others, 1997). More recent numerical work shows that anisotropy and convergent flow can produce folds spanning scales from cloudy bands to radar reflectors (Bons, Jansen, and others, 2016; Bons, Hu, and others, 2025; Zhang and others, 2024). The analysis below concerns passive age transport in a prescribed velocity field. It supplies a necessary kinematic test for folding, not a dispersion relation for active instability.

2.2 The WBA local wrinkle criterion

Let m be the slope of an infinitesimal material-line segment in a two-dimensional velocity field (u, w) . WBA derived

$$\frac{Dm}{Dt} = w_x + (w_z - u_x)m - u_z m^2. \quad (1)$$

Equation (1) is the Riccati form of the material-line equation and is standard continuum kinematics. WBA's distinctive step was to compare the rotation of a disturbed tangent with the rotation of the local steady isochrone tangent m_0 . A second equal-rotation direction separates disturbances that initially flatten toward m_0 from those that rotate away from it. Their shear number expresses the competition between shear-driven rotation and longitudinal extension.

Equation (1) is local. Applied pointwise along a complete curve, it contains the exact tangent kinematics of that curve. The published WBA stability diagnostic, however, does not by itself determine whether an entire equal-age surface loses its horizontal ordering. WBA emphasized this limitation over rough topography: a disturbed tangent can open relative to the local steady isochrone on the lee side of a bump while merely recovering deformation accumulated on the stoss side (Waddington and others, 2001).

2.3 Finite strain and steady isochrone geometry

Jacobson and Waddington (2004) integrated finite deformation along particle paths rather than relying on a homogeneous instantaneous velocity gradient. This is the appropriate framework for deciding whether an inherited open fold becomes recumbent after a long and spatially variable strain history. Their later divide-migration calculation supplied a concrete source of the inherited perturbation (Jacobson and Waddington, 2005).

A separate line of analysis concerns the geometry of isochrones generated continuously in a steady flow. Parrenin, Hindmarsh, and Rémy (2006) introduced logarithmic total-flux and normalized-streamfunction coordinates in which particle trajectories are straight characteristics when the vertical velocity shape is horizontally uniform. Hindmarsh, Leysinger Vieli, Raymond, and others (2006) compared streamlines and isochrones over basal topography and showed that they are close when

$$\frac{\dot{a}}{U} \frac{L}{H} \ll 1, \quad (2)$$

where $\dot{a}$ is accumulation, U a horizontal speed, L the forcing wavelength and H the ice thickness. Parrenin and Hindmarsh (2007) generalized the characteristic analysis to horizontally non-uniform velocity profiles and wrote isochrone slope as an instantaneous normalized-streamfunction contribution plus a path-history term. In their notation, the sign of $1 - \alpha$ determines whether neighboring particles preserve their order, and $\alpha > 1$ corresponds to an isochrone that has passed through vertical. They proved that this cannot occur if horizontal velocity increases upward. Hindmarsh, Leysinger Vieli, and Parrenin (2009) used these solutions to verify a large-scale numerical age model, and Parrenin, Chung, and others (2025) recently implemented the same flux-coordinate framework for age, particle origin and vertical thinning.

Table 1 locates the present analysis within this literature.

3 Steady age transport and the order Jacobian

3.1 Setting and assumptions

Consider a steady two-dimensional glacier velocity

$$\mathbf{u}(x, z) = (u(x, z), w(x, z)) \quad (3)$$

in a domain bounded above by the steady surface $z = s(x)$. The kinematic result requires only the following assumptions on the region and trajectories of interest:

(A1) The velocity is continuously differentiable and incompressible,

$$u_x + w_z = 0. \quad (4)$$

Table 1: Relationship between established folding analyses and the present formulation.

Study	Primary question	Diagnostic and scope
Waddington and others (2001)	Does a pre-existing local wrinkle flatten or rotate toward overturning?	Instantaneous tangent slope relative to a steady reference isochrone.
Jacobson and Waddington (2004)	Does a finite-strain history overturn an injected disturbance?	Deformation gradient integrated along trajectories.
Parrenin, Hindmarsh, and Rémy (2006) and Parrenin and Hindmarsh (2007)	How does steady flow shape isochrones and preserve particle order?	Logarithmic flux coordinates, characteristic age solutions and a path factor; monotonic upward speed precludes vertical isochrones.
Hindmarsh, Leysinger Vieli, Raymond, and others (2006)	Do layers drape or override bed and basal-boundary perturbations?	Perturbation and full-mechanical models; wavelength controls architecture.
Hindmarsh, Leysinger Vieli, and Parrenin (2009) and Parrenin, Chung, and others (2025)	How can age and thinning be computed accurately in steady flow tubes?	Eulerian and characteristic solvers for age, origin and vertical thinning.
Gudmundsson (1997a)	Can rough-bed Stokes flow contain local extrusion?	Analytical velocity extrema and reverse vertical shear near sinusoidal relief.
This study	Has cumulative reverse shear erased depositional ordering?	Signed inverse-thinning factor, positive–negative shear budget, full-Stokes threshold, obstacle bounds and trajectory diagnostic.

(A2) Horizontal motion is one-way,

$$u(x, z) > 0, \quad (5)$$

after orienting x downstream.

160 (A3) Every trajectory considered enters the ice through a unique accumulating branch of the surface. The surface accumulation rate, expressed in vertical ice-equivalent units, satisfies $\dot{a}(x) > 0$ on that branch.

(A4) Stratigraphic age is a passive, non-diffusive scalar. Its steady value $\tau(x, z)$ satisfies

$$u\tau_x + w\tau_z = 1, \quad \tau = 0 \quad \text{at deposition.} \quad (6)$$

165 An impermeable bed is not required for the local result. Basal melting may terminate some trajectories without altering the derivation before termination. Basal freezing, in contrast, introduces a second material source and requires separate origin labels. Assumption (A2) excludes horizontal stagnation and closed circulation. Assumption (A3) requires special care in a blue-ice area because the local surface is an outlet; the material origin must instead be traced to
170 an upstream inflow branch or another source.

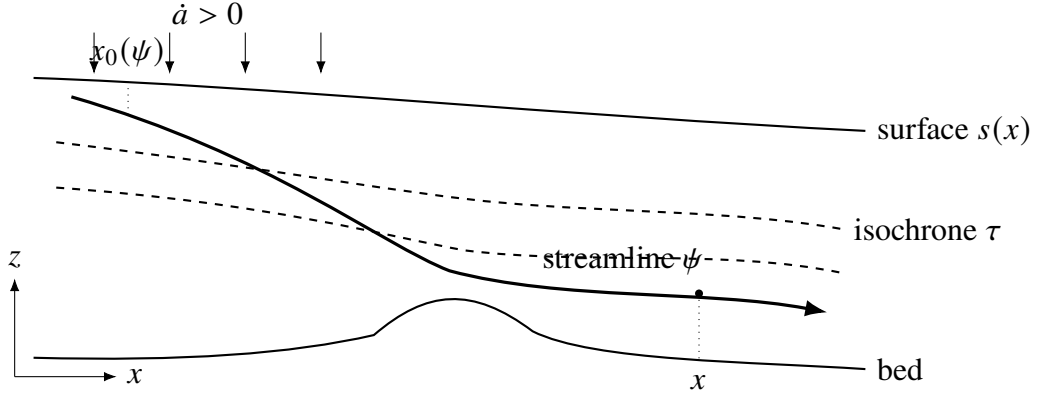


Figure 1: Steady plane-flow geometry. Each streamline considered here enters through a unique accumulating surface point $x_0(\psi)$. The order Jacobian compares the downstream positions, at equal age, of particles deposited at neighboring values of x_0 . A bed obstacle may curve isochrones without reversing their horizontal order.

Because of Equation (4), there is a streamfunction ψ such that

$$u = \psi_z, \quad w = -\psi_x. \quad (7)$$

Combining Equations (6) and (7) gives the exact Jacobian identity

$$\frac{\partial(\tau, \psi)}{\partial(x, z)} = \tau_x \psi_z - \tau_z \psi_x = u \tau_x + w \tau_z = 1. \quad (8)$$

175 Thus age and streamfunction form local area-preserving coordinates. Isochrones and streamlines intersect transversely wherever the assumptions hold. This is a classical flow-coordinate construction; it is useful here because its inverse directly measures stratigraphic order.

3.2 Surface origin and travel time

The streamfunction restricted to the surface is

$$180 \quad Q(x) = \psi(x, s(x)). \quad (9)$$

The steady surface kinematic condition is

$$u_s s_x - w_s = \dot{a}, \quad (10)$$

where the subscript s denotes evaluation at the surface. Therefore

$$Q'(x) = \psi_x + \psi_z s_x = -w_s + u_s s_x = \dot{a}(x) > 0. \quad (11)$$

185 Every streamline label ψ has a unique deposition position $x_0(\psi)$ satisfying

$$Q(x_0) = \psi, \quad \frac{dx_0}{d\psi} = \frac{1}{\dot{a}_0}, \quad (12)$$

where $\dot{a}_0 = \dot{a}(x_0)$.

At fixed ψ , Equation (6) gives

$$\left. \frac{\partial \tau}{\partial x} \right|_{\psi} = \frac{1}{u}. \quad (13)$$

190 Consequently the age at (x, ψ) is the travel time from deposition,

$$\tau(x, \psi) = \int_{x_0(\psi)}^x \frac{d\xi}{u(\xi, z(\xi, \psi))}. \quad (14)$$

At fixed x , Equation (7) implies

$$z_{,\psi} = \frac{1}{u}. \quad (15)$$

Differentiating Equation (14) with respect to ψ , using Equations (12) and (15), yields

$$195 \quad \tau_{\psi} = -\frac{1}{\dot{a}_0 u_{s0}} - \int_{x_0}^x \frac{u_z}{u^3} d\xi, \quad (16)$$

where $u_{s0} = u(x_0, s(x_0))$. The first term records the separation imposed at deposition; the integral records the subsequent differential travel time between neighboring streamlines.

3.3 Exact order theorem

Parameterize an isochrone of fixed age by the surface deposition coordinate x_0 . The relevant
200 quantity is not its curvature but the horizontal projection of its tangent.

Definition 1 (Isochrone-order Jacobian). The isochrone-order Jacobian is

$$\mathcal{J}(x, x_0) = \left. \frac{\partial x}{\partial x_0} \right|_{\tau}. \quad (17)$$

It measures whether particles deposited in increasing order of x_0 remain in increasing downstream order on an equal-age surface.

205 **Theorem 1** (Cumulative order criterion). *Under assumptions (A1)–(A4), the isochrone-order Jacobian is*

$$\boxed{\mathcal{J} = \frac{u(x, z)}{u_{s0}} \left[1 + \dot{a}_0 u_{s0} \int_{x_0}^x \frac{u_z}{u^3} d\xi \right]}. \quad (18)$$

It is also related exactly to the vertical age gradient by

$$\boxed{\mathcal{J} = -\dot{a}_0 \tau_z}. \quad (19)$$

210 Hence $\mathcal{J} > 0$ denotes ordered stratigraphy, $\mathcal{J} = 0$ a vertical tangent, and $\mathcal{J} < 0$ an overturned limb.

Proof. At fixed age, differentiation of $\tau(x, \psi)$ gives

$$\left. \frac{\partial x}{\partial \psi} \right|_{\tau} = -\frac{\tau_{\psi}}{\tau_x|_{\psi}} = -u\tau_{\psi}. \quad (20)$$

Multiplication by $d\psi/dx_0 = \dot{a}_0$ and substitution of Equation (16) gives Equation (18). At fixed x ,
 215 $\tau_z = \tau_{\psi}\psi_z = u\tau_{\psi}$, which gives Equation (19). $\square$

The vertical thinning function λ_z is conventionally the present vertical thickness of an infinitesimal age layer divided by its ice-equivalent thickness at deposition. For an ordered layer, a positive age increment $d\tau$ corresponds to the present vertical thickness $-dz = -d\tau/\tau_z$ and the initial thickness $\dot{a}_0 d\tau$. Hence

$$220 \quad \lambda_z = -\frac{1}{\dot{a}_0\tau_z} = \mathcal{J}^{-1}, \quad \mathcal{J} > 0. \quad (21)$$

Thus $\mathcal{J}$ is not a new independent state variable: in the ordered regime it is the inverse of the established vertical thinning function (Parrenin, 2013; Parrenin, Chung, and others, 2025). Equation (19) provides its signed continuation through a vertical tangent. The standard positive thinning function diverges as $\tau_z \rightarrow 0$, whereas $\mathcal{J}$ crosses zero regularly and records orientation
 225 reversal.

At deposition, $x = x_0$, $u = u_{s0}$, and $\mathcal{J} = 1$. At the first vertical tangent, $\tau_z = 0$, but the material curve remains regular. Inverting Equation (8) shows that

$$\left. \frac{\partial z}{\partial x_0} \right|_{\tau} = \dot{a}_0\tau_x. \quad (22)$$

When $\mathcal{J} = 0$, Equation (6) gives $\tau_x = 1/u$, and therefore

$$230 \quad \left. \frac{\partial z}{\partial x_0} \right|_{\tau, \mathcal{J}=0} = \frac{\dot{a}_0}{u} > 0. \quad (23)$$

The divergence of the conventional slope $m = dz/dx$ at overturning is therefore a coordinate singularity, not a singularity of the layer itself.

3.4 Positive and reverse shear budgets

Define

$$235 \quad u_z^+ = \max(u_z, 0), \quad u_z^- = \max(-u_z, 0), \quad (24)$$

and the dimensionless cumulative contributions

$$\mathcal{P}(x, x_0) = \dot{a}_0 u_{s0} \int_{x_0}^x \frac{u_z^+}{u^3} d\xi, \quad (25)$$

$$\mathcal{R}(x, x_0) = \dot{a}_0 u_{s0} \int_{x_0}^x \frac{u_z^-}{u^3} d\xi. \quad (26)$$

Equation (18) becomes

$$\mathcal{J} = \frac{u}{u_{s0}} (1 + \mathcal{P} - \mathcal{R}). \quad (27)$$

Because $u/u_{s0} > 0$, the necessary-and-sufficient condition for preservation of horizontal order is

$$\mathcal{R} < 1 + \mathcal{P}. \quad (28)$$

The first vertical tangent occurs at

$$\mathcal{R} = 1 + \mathcal{P}, \quad (29)$$

and overturning occurs when $\mathcal{R} > 1 + \mathcal{P}$.

The unit term in Equations (27)–(29) is the finite initial ordering imposed at deposition. Positive u_z , the usual surface-fast glacier profile, strengthens that ordering. Reverse shear must overcome both. Thus a local region with $u_z < 0$ is not itself a fold criterion.

In travel-time coordinates, $dx = u dt$, so

$$\mathcal{R} = \dot{a}_0 u_{s0} \int_0^T \frac{u_z^-}{u^2} dt. \quad (30)$$

Reverse shear is amplified by residence time and by slow horizontal motion. This weighting explains why local extrusion in fast throughflow may be harmless, while modest reverse shear near stagnation can dominate the order budget.

Corollary 1 (Monotonic-shear result). *If $u_z \geq 0$ everywhere along a trajectory, then $\mathcal{R} = 0$, $\mathcal{J} > 0$, and the steady isochrone cannot overturn.*

This corollary is the physical-coordinate statement of the no-overturning result in Appendix A of Parrenin and Hindmarsh (2007). It is sufficient, not necessary. Equation (28) permits finite intervals of reverse shear provided their cumulative weighted contribution remains subcritical.

3.5 Relation to the Parrenin–Hindmarsh path factor

Parrenin and Hindmarsh (2007) used a normalized streamfunction and logarithmic flux coordinates. For two neighboring particles, their Appendix A gives a mapping of the form

$$\delta\pi = (1 - \alpha) \delta\pi_0, \quad (31)$$

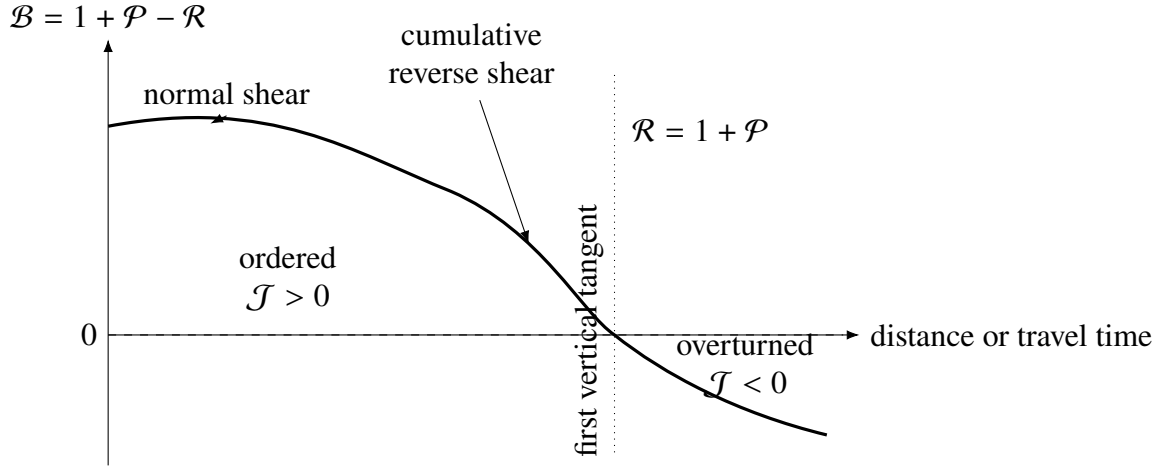


Figure 2: Conceptual order budget. The sign of $\mathcal{B} = 1 + \mathcal{P} - \mathcal{R}$ is the sign of the order Jacobian. Local reverse shear need not cause overturning; a vertical tangent appears only after the cumulative reverse-shear budget reaches the unit depositional margin plus accumulated normal shear.

where $\alpha > 1$ corresponds to an isochrone exceeding vertical. Their present and surface-origin coordinates are monotone functions of x and x_0 wherever assumptions (A2)–(A3) hold.

Consequently,

$$\text{sgn}(1 - \alpha) = \text{sgn}\left(\left.\frac{\partial x}{\partial x_0}\right|_{\tau}\right) = \text{sgn}(\mathcal{J}) = \text{sgn}(-\tau_z). \quad (32)$$

The present result is therefore not an independent no-overturning theorem. Nor is $\mathcal{J}$ independent of the established thinning variable: Equation (21) identifies it as the signed inverse vertical-thinning factor. The contribution of the physical-space representation is that it exposes the positive and negative parts of vertical shear, remains regular at the first vertical tangent and can be evaluated directly from a numerical velocity field. The current `age_flow_line` formulation assumes strictly positive accumulation for its horizontal flux coordinate and no reverse horizontal flow for its vertical flux coordinate; it computes age and positive vertical thinning within that admissible regime (Parrenin, Chung, and others, 2025).

4 What does full-Stokes mechanics imply?

4.1 Exact stress decomposition of vertical shear

The order theorem is kinematic. We now ask whether ordinary glacier mechanics forces Equation (28) to hold. Consider steady, isothermal or thermomechanically prescribed, isotropic

generalized-Newtonian Stokes flow,

$$\nabla \cdot \mathbf{u} = 0, \quad (33)$$

$$-\nabla p + \nabla \cdot \boldsymbol{\tau} + \rho \mathbf{g} = \mathbf{0}, \quad (34)$$

$$\boldsymbol{\tau} = 2\eta \mathbf{D}, \quad \mathbf{D} = \frac{1}{2} (\nabla \mathbf{u} + \nabla \mathbf{u}^\top), \quad \eta > 0. \quad (35)$$

The scalar viscosity η may depend on temperature and effective strain rate. Anisotropic rheology is discussed separately below.

With z vertical and gravity having no horizontal component, the horizontal momentum equation is

$$-p_x + \tau_{xx,x} + \tau_{xz,z} = 0. \quad (36)$$

Write

$$p = p_{\text{atm}} + \rho g(s - z) + p_{\text{nh}}, \quad (37)$$

where p_{nh} contains all nonhydrostatic pressure, including the finite-slope surface correction. Integrating Equation (36) downward from the free surface gives

$$\tau_{xz}(x, z) = \tau_{xz}^s - \rho g s_x(s - z) + \int_z^s (\tau_{xx,x} - p_{\text{nh},x}) \, d\zeta. \quad (38)$$

Because $\tau_{xz} = \eta(u_z + w_x)$,

$$u_z = \underbrace{\frac{\rho g(s - z)(-s_x)}{\eta}}_G + \underbrace{\left[-w_x + \frac{\tau_{xz}^s}{\eta} + \frac{1}{\eta} \int_z^s (\tau_{xx,x} - p_{\text{nh},x}) \, d\zeta \right]}_{R_{\text{FS}}}. \quad (39)$$

For a surface descending downstream, $s_x \leq 0$, the gravity-driven term G is nonnegative and favors order preservation. The residual R_{FS} contains the full-Stokes effects omitted from the shallow-ice shear balance: gradients of vertical velocity, longitudinal stress, nonhydrostatic pressure and the Cartesian surface shear-stress component.

Substitution into Equation (18) gives

$$\mathcal{J} = \frac{u}{u_{s0}} (1 + \mathcal{G} + \mathcal{F}), \quad (40)$$

with

$$\mathcal{G} = \dot{a}_0 u_{s0} \int_{x_0}^x \frac{G}{u^3} \, d\xi \geq 0, \quad (41)$$

$$\mathcal{F} = \dot{a}_0 u_{s0} \int_{x_0}^x \frac{R_{\text{FS}}}{u^3} \, d\xi. \quad (42)$$

The exact full-Stokes failure condition is

$$\mathcal{F} \leq -(1 + \mathcal{G}). \quad (43)$$

A conservative sufficient condition for no overturning is

$$\dot{a}_0 u_{s0} \int_{x_0}^x \frac{[R_{FS}]_-}{u^3} d\xi < 1. \quad (44)$$

This inequality discards all positive contributions, including gravity-driven shear, and is therefore stronger than necessary but easy to test.

4.2 There is no universal full-Stokes sign theorem

Neither incompressibility nor the Stokes equations fixes the sign of u_z . The constitutive relation constrains $u_z + w_x$, while longitudinal stress and nonhydrostatic pressure can alter the Cartesian vertical gradient of horizontal speed. Analytical rough-bed solutions demonstrate local velocity maxima and minima, and therefore regions with $u_z < 0$, even for gravity-driven linear viscous flow (Gudmundsson, 1997a). The historical extrusion-flow literature likewise shows that surface-fast velocity profiles are not a pointwise theorem of full-Stokes mechanics (Waddington, 2010).

At stronger roughness, the one-way-flow assumption itself can fail. Gudmundsson (1997b) found separated secondary circulation in troughs of sufficiently steep sinusoidal beds. Ideal no-slip Stokes flow near a sharp corner can contain nested Moffatt eddies (Moffatt, 1964). These examples are not direct models of Allan Hills: glacier beds may slide, Glen flow is nonlinear, and real corners are rounded. Their theoretical significance is that a steady creeping-flow equation does not forbid recirculation. When $u = 0$ or trajectories close, the downstream travel-time coordinate and unique surface-origin map cease to exist, so Equations (18)–(28) no longer apply globally.

The distinction between local extrusion and cumulative order reversal is central. Published rough-bed solutions establish that u_z can be negative. They do not, by that fact alone, establish that $\mathcal{R}$ reaches $1 + \mathcal{P}$. The latter is an order-one integrated threshold.

5 Regimes that preserve order

5.1 Shallow-ice flow

Under the conventional shallow-ice approximation,

$$\tau_{xz} \simeq -\rho g(s - z)s_x. \quad (45)$$

For downstream flow with $s_x < 0$, $\tau_{xz} > 0$. An isotropic Glen-type rheology gives u_z with the same sign as τ_{xz} , while leading-order basal sliding contributes a depth-independent horizontal speed (Cuffey and Paterson, 2010). Therefore

$$u_z \geq 0, \quad \mathcal{R} = 0, \quad \mathcal{J} > 0. \quad (46)$$

A fixed bed obstacle can produce arches, troughs and finite curvature in shallow-ice stratigraphy, but it cannot overturn a steady isochrone under these assumptions.

5.2 Regular thin-film full-Stokes flow

Let

$$\varepsilon = \frac{H}{L} \ll 1 \quad (47)$$

be the aspect ratio of a smooth glacier. Higher-order glacier models reproduce the Stokes solution to second order in ε under their regular asymptotic assumptions, including a range of basal slip regimes (Blatter, 1995; Schoof and Hindmarsh, 2010). Suppose the non-shallow contribution scales as

$$R_{\text{FS}} = O\left(\varepsilon^2 \frac{U}{H}\right) \quad (48)$$

through a region of length $L = H/\varepsilon$, and suppose u remains comparable to U . Then

$$\mathcal{F} = O\left(\frac{\dot{a}}{U} \varepsilon\right). \quad (49)$$

Thus small regular full-Stokes corrections cannot erase the unit depositional margin when

$$\frac{\dot{a}}{U} \varepsilon \ll 1. \quad (50)$$

This is an asymptotic sufficient regime, not a uniform theorem. It can fail near divides, abrupt sliding transitions, contact zones, steep obstacles, very slow flow or other regions where the thin-film expansion is singular or nonuniform.

5.3 A sufficient bound for a localized obstacle

The cumulative criterion yields a direct obstacle bound without requiring pointwise monotonic shear.

Proposition 1 (Localized reverse-shear bound). *Suppose reverse shear is confined to a trajectory segment of horizontal length ℓ_- . On that segment, assume*

$$u \geq rU, \quad u_{s0} \leq r_s U, \quad u_z^- \leq C \frac{U}{d}, \quad (51)$$

where $U > 0$ is a representative speed, d is the vertical deformation scale, $r > 0$, $r_s > 0$, and $C > 0$. Then

$$\mathcal{R} \leq C \frac{r_s}{r^3} \frac{\dot{a}_0}{U} \frac{\ell_-}{d}. \quad (52)$$

A sufficient condition for no overturning is that the right-hand side be less than one.

Proof. Substitute Equation (51) into Equation (26) and bound the integral by its maximum integrand multiplied by ℓ_- . $\square$

For an isolated obstacle whose horizontal and vertical influence scales are both $O(H)$, with non-stagnant throughflow,

$$\mathcal{R} = O\left(\frac{\dot{a}}{U}\right). \quad (53)$$

In a typical interior or flank setting where $U \gg \dot{a}$, an isolated fixed bump can create local extrusion and visible draping but is far from the order-one reverse-shear dose required for overturning. Conversely, a long reverse-shear region with $d \sim H$ approaches the threshold only when

$$\frac{\ell_-}{H} \gtrsim \frac{U}{\dot{a}_0} \frac{r^3}{Cr_s}, \quad (54)$$

unless the flow becomes slow enough for the r^{-3} factor to dominate.

Equation (52) makes precise why “flow over a bump cannot fold an isochrone” is often a good glacier approximation but not a universal Stokes theorem. The approximation depends on non-stagnant flow, a localized deformation zone and a small accumulation-to-speed ratio.

5.4 First-order cancellation over periodic roughness about plug flow

A second reason that fixed roughness may have weak cumulative effect is cancellation, but the clean result requires a plug-flow background. Let

$$u(x, z; \epsilon_b) = U_b + \epsilon_b u_1(x, z) + O(\epsilon_b^2), \quad U_b > 0, \quad (55)$$

where U_b is spatially uniform and u_1 is periodic in x , with wavelength $2\pi/k$ and zero horizontal mean at each elevation. Along the unperturbed trajectory $z = z_0$,

$$\frac{u_z}{u^3} = \epsilon_b \frac{u_{1,z}(x, z_0)}{U_b^3} + O(\epsilon_b^2). \quad (56)$$

Therefore, over one complete wavelength,

$$\int_x^{x+2\pi/k} \frac{u_z}{u^3} d\xi = O(\epsilon_b^2). \quad (57)$$

The perturbative frictionless rough-bed solution of Gudmundsson (1997a) has this plug-flow structure: local extrusion occurs, but its first-order stoss- and lee-side contributions cancel over a

complete symmetric wavelength.

Cancellation is not guaranteed for a sheared background, a solitary asymmetric obstacle,
 390 a finite path ending mid-cycle, strongly nonlinear roughness or near-stagnant trajectories.
 Equation (57) is therefore a restricted asymptotic result, not a general symmetry theorem.

6 How the order theorem can fail

6.1 Cumulative extrusion without stagnation

The most direct steady-flow route is Equation (29). A glacier with $u > 0$ can overturn a steady
 395 isochrone if reverse shear is sufficiently strong, sufficiently persistent or sufficiently concentrated
 where u is small. The exact threshold is

$$\dot{a}_0 u_{s0} \int_{x_0}^x \frac{u_z^-}{u^3} d\xi = 1 + \dot{a}_0 u_{s0} \int_{x_0}^x \frac{u_z^+}{u^3} d\xi. \quad (58)$$

No change in boundary conditions is logically necessary if a steady full-Stokes velocity field
 satisfies this equality. The existing extrusion-flow literature establishes plausible ingredients but,
 400 to our knowledge, has not evaluated this cumulative isochrone-order threshold.

6.2 Stagnation, separation and closed circulation

If u reaches zero, the x -travel-time representation becomes singular. If a secondary circulation
 forms, a streamline may not have a unique surface origin and may intersect an equal-age surface
 in a topology not represented by a single downstream coordinate. Strong roughness, cavities
 405 and sharp corners are therefore not merely large values of $\mathcal{R}$; they can invalidate the coordinate
 system on which the theorem rests. Numerical calculations should treat $u = 0$ as an event and
 switch to a more general Lagrangian deformation analysis.

6.3 Surface ablation and sublimation

An accumulating surface is an inflow boundary on which newly deposited material has $\tau = 0$.
 410 An ablating surface is an outflow boundary at which existing isochrones terminate. In a blue-ice
 area, the surface streamfunction need not be globally monotone because

$$Q'(x) = \dot{a}(x) \quad (59)$$

changes sign between accumulation and ablation zones. A streamline may enter through an
 upstream accumulation region and leave through a downstream ablation region. The theorem
 415 can still be applied to an interior trajectory if that trajectory has a unique origin on an upstream
 accumulating branch, but Q^{-1} must be constructed on that branch rather than on the entire

local surface. Multiple inflow branches or an origin on basal freeze-on require a more general material-source label.

Steady ablation most naturally produces upward-bent and truncated layers, not an intact fold. However, migration, onset or cessation of an ablation zone changes the age field relative to the new steady solution and supplies precisely the inherited disturbance required by WBA. Surface sublimation followed by burial can also preserve an unconformity with steep or truncated limbs. At Allan Hills, the present surface is an ablation boundary and therefore lies outside the simplest theorem even if the subsurface velocity is steady.

6.4 Basal melting and freeze-on

Basal melting removes trajectories and truncates layers. Basal freeze-on introduces ice through a second inflow boundary, so one surface deposition coordinate cannot label the entire stratigraphy. Numerical models show that freeze-on can generate plume-like structures and complex internal layering extending through a large fraction of the ice thickness (Leysinger Vieli and others, 2018). A moving melt/freeze boundary additionally makes the velocity and age problem time dependent.

6.5 Time-dependent geometry and basal traction

If the bed, free surface, divide position or basal traction changes, the inherited age field is not the steady age solution of the new velocity field. The perturbation then obeys a homogeneous advection equation only after the change; it need not vanish initially. Divide migration converts a formerly steady Raymond arch into a fold seed (Jacobson and Waddington, 2005). Traveling slippery or sticky patches produce moving-frame critical layers and thickness-scale overturning (Wolovick, Creyts, and others, 2014; Wolovick and Creyts, 2016). Bedrock uplift, rotation or erosion similarly changes the reference stratigraphy and can inject a finite-amplitude disturbance even when subsequent flow is steady.

6.6 Active rheology, anisotropy and three-dimensional flow

Equation (39) assumes isotropic scalar viscosity. Crystal fabric, impurities, grain size and damage can alter both the stress–strain relation and the velocity field in response to stratigraphy. Full-Stokes anisotropic divide models already demonstrate substantial changes in isochrone shape and age–depth relations (Martín, Gudmundsson, and others, 2009). More strongly coupled heterogeneity can generate folds internally rather than merely advecting an imposed perturbation (Alley and others, 1997; Thorsteinsson and Waddington, 2002; Bons, Jansen, and others, 2016; Bons, Hu, and others, 2025). The Allan Hills microstructure indicates that anisotropy and localized shear cannot be ignored (Stoll and others, 2026).

The order theorem is also intrinsically two-dimensional. In three dimensions, incompressibility provides a vector potential rather than a scalar streamfunction, and neighboring surface origins

can exchange order in one horizontal projection while remaining separated in the other. Divide models show that along-ridge flow and divide migration alter isochrone architecture even when cross-flow is modest (Martín, Hindmarsh, and others, 2009). A three-dimensional generalization should track the two-dimensional Jacobian mapping a surface deposition patch to an equal-age surface. The scalar $\mathcal{J}$ is recovered only when cross-flow is negligible and a coherent flow tube exists.

7 Implications for Allan Hills and COLDEX

The Allan Hills setting sits close to several limits in which a fixed-obstacle argument becomes unreliable. Present-day horizontal motion near ALHIC1901 is only centimetres per year, the ice rises toward an ablating surface, and the bed contains steep buried nunatak geometry (Spaulding and others, 2012; Kehrl and others, 2018; Stoll and others, 2026). Slow motion enhances the u^{-3} weighting in Equation (26). The local ablation surface is an outlet rather than the deposition boundary, so relevant source-to-core trajectories must be traced to an upstream accumulation branch or another material source. Strong relief raises the possibility of nonhydrostatic stress, local extrusion and near-stagnant flow. The small parameter $\dot{a}/U$ that protects ordinary fast flank flow may therefore not be small along the relevant source-to-core trajectories.

At the same time, the observed deformation cannot be attributed to present-day steady geometry without a trajectory calculation. Steep dip is not overturning: the 69° mean dip in ALHIC2302 remains below vertical (Kirkpatrick and others, 2025). Centimetre-scale z-folds in ALHIC1901 demonstrate local folding but are too small by themselves to explain all core-scale age reversals (Stoll and others, 2026). The inverted age–depth relation in ALHIC1901 is the more direct evidence that τ_z has changed sign along at least some sampling intervals (Shackleton and others, 2025). Equation (19) translates that observation into $\mathcal{J} < 0$ if the sampled layers belong to a single surface-origin family; alternatively, multiple source boundaries, unconformities or faults could also generate apparent age reversal.

The theory suggests a falsifiable hierarchy of hypotheses.

- (1) **Steady fixed geometry.** Compute the present or reconstructed steady full-Stokes field and evaluate $\mathcal{P}$, $\mathcal{R}$ and $\mathcal{J}$ along source-to-core trajectories. If $1 + \mathcal{P} - \mathcal{R}$ remains positive with a substantial margin, fixed present-day geometry cannot produce the observed recumbent folding.
- (2) **Steady but singular or multi-source flow.** Test for stagnation, recirculation, basal freeze-on and surface origin multiplicity. If these occur, the simple order theorem is inapplicable and the full material map must be used.
- (3) **Inherited unsteady stratigraphy.** If the steady model preserves order, evaluate changes in ablation pattern, ice thickness, flow direction, divide position, bed geometry and basal traction. These mechanisms inject finite disturbances that can later be amplified by shear.

(4) **Active deformation.** If kinematic models underpredict observed folds, couple fabric and rheological heterogeneity to flow. The microstructural evidence makes this a likely contributor at least at centimetre scale.

This hierarchy sharpens, rather than proves, the intuition that boundary-condition changes are required. They are effectively required in the regular shallow, non-stagnant regime because Equation (52) leaves a large order margin. They are not mathematically required for every full-Stokes glacier: sufficiently cumulative extrusion or loss of unique throughflow can overturn steady isochrones.

8 Numerical diagnostic for full-Stokes models

The order criterion can be evaluated without explicitly solving the age equation or advecting a dense family of isochrones. Release trajectories from an accumulating inflow boundary and record x_0 , $\dot{a}_0$ and u_{s0} . Along each trajectory integrate

$$\frac{dx}{dt} = u, \quad \frac{dz}{dt} = w, \quad (60)$$

$$\frac{d\mathcal{P}}{dt} = \dot{a}_0 u_{s0} \frac{u_z^+}{u^2}, \quad \frac{d\mathcal{R}}{dt} = \dot{a}_0 u_{s0} \frac{u_z^-}{u^2}, \quad (61)$$

with $\mathcal{P}(0) = \mathcal{R}(0) = 0$. Then

$$\mathcal{J}(t) = \frac{u(t)}{u_{s0}} [1 + \mathcal{P}(t) - \mathcal{R}(t)]. \quad (62)$$

The diagnostic is

$$\min_t [1 + \mathcal{P}(t) - \mathcal{R}(t)] \begin{cases} > 0, & \text{order preserved,} \\ = 0, & \text{first vertical tangent,} \\ < 0, & \text{overturned steady isochrone.} \end{cases} \quad (63)$$

A robust implementation should include four checks. First, compare Equation (62) with $-\dot{a}_0 \tau_z$ from an independently solved steady age equation. Second, advect at least one pair of neighboring particles to verify the sign of $\partial x / \partial x_0|_\tau$. Third, stop the downstream-coordinate integration when u approaches zero and diagnose recirculation with the full trajectory topology. Fourth, compute u_z using the finite-element gradient rather than differencing noisy interpolated velocities. Separate storage of $\mathcal{P}$ and $\mathcal{R}$ is preferable to integrating their difference because large compensating contributions can otherwise obscure numerical error.

For transient flow, Equation (62) is not valid because the steady travel-time representation fails. The corresponding calculation should integrate the deformation of the two-parameter

deposition map through time. Nevertheless, the steady order budget remains useful as a baseline
515 and as a diagnostic applied to quasi-steady snapshots.

9 Discussion

9.1 A theorem about deposition order, not about curvature

The curvature equation for an advected material line contains first and second spatial derivatives
of velocity. It correctly shows that an obstacle can create curvature from an initially straight
520 layer. The present order criterion answers a different question. A recumbent fold requires the
horizontal projection of an equal-age surface to reverse, or equivalently τ_z to change sign. An
obstacle may generate large curvature while $\mathcal{J}$ remains positive everywhere. This is the rigorous
sense in which a steady bump can disturb isochrones without supplying the order-reversing seed
required for a recumbent fold.

525 WBA's statement that steady-state layers are not passively folded can now be made conditional.
If a unique steady age field exists and Equation (28) holds, exact steady stratigraphy is an invariant
solution and remains ordered. A prescribed steady flow cannot spontaneously create a perturbation
relative to that solution. However, arbitrary material lines placed in the same velocity field are
not subject to this restriction, and the restriction can fail if cumulative reverse shear crosses the
530 threshold or if the surface-origin coordinate ceases to be global.

9.2 Why local extrusion is usually insufficient

The full-Stokes literature demonstrates local extrusion but does not imply order reversal. Three
effects protect the isochrone. First, deposition provides a finite unit margin in Equation (27).
Second, ordinary surface-fast shear adds $\mathcal{P}$. Third, reverse shear around a localized or periodic
535 obstacle is short-lived and can cancel between stoss and lee sides. The bound in Equation (52)
quantifies the first two effects, while Equation (57) quantifies the third in a perturbative setting.

Near stagnation the protection weakens sharply. The u^{-3} spatial weighting and u^{-2} temporal
weighting imply that a slowly moving basal zone can accumulate a large order change even when
the pointwise shear is moderate. This is also the regime in which age becomes large, numerical
540 conditioning deteriorates and recirculation becomes possible. A convincing steady-obstacle
explanation for Allan Hills folding should therefore identify a specific slow reverse-shear pathway
and demonstrate that it crosses Equation (29); showing $u_z < 0$ is insufficient.

9.3 Boundary changes as fold seeds

Time-dependent boundary processes remain the most natural route to finite inherited disturbances.
545 Surface sublimation can truncate and steepen layers; migration of the ablation boundary can
leave that geometry to be buried or sheared. Basal melt and freeze-on alter source and sink

topology. Divide migration changes the reference velocity profile. Traveling traction anomalies create a moving-frame critical layer and material trapping. Bedrock uplift or rotation changes both geometry and stress. Each mechanism breaks at least one assumption behind the steady order theorem and can therefore create a nonzero perturbation relative to the current steady stratigraphy.

The order framework does not identify which mechanism operated at Allan Hills. It supplies a rejection test. If a high-resolution, history-consistent full-Stokes model of fixed geometry satisfies $\mathcal{J} > 0$ by a wide margin on all relevant trajectories, then steady topographic draping is excluded as the cause of recumbent age reversals. The remaining explanation must invoke time dependence, multiple material sources, active rheology, three-dimensional exchange or structural discontinuity.

9.4 Novelty and relationship to established theory

Several components of this manuscript are established. Equation (8) is a standard consequence of steady incompressible transport. Logarithmic flux coordinates and characteristic age solutions were developed by Parrenin, Hindmarsh, and Rémy (2006) and Parrenin and Hindmarsh (2007), and the latter paper contains both the monotonic-shear corollary and the path dependence of isochrone slope. The vertical thinning function was derived in the same framework by Parrenin (2013) and is now computed operationally by Parrenin, Chung, and others (2025); Equation (21) shows that the present order factor is its inverse in the ordered regime. The need for a transient wrinkle and the competition between shear and extension are due to Waddington and others (2001). Finite-strain amplification is treated by Jacobson and Waddington (2004). Local extrusion and rough-bed recirculation are known full-Stokes phenomena (Gudmundsson, 1997a; Gudmundsson, 1997b).

The specific analytical contribution is therefore not the age–streamfunction coordinate system, the thinning factor or the monotonic-shear theorem. It is the explicit physical-space identity that writes the signed inverse thinning as a cumulative u_z/u^3 integral, its positive–negative decomposition in Equations (25)–(28), and the mechanical consequences drawn from that decomposition. A targeted search of the steady-isochrone, age-model and extrusion-flow literature did not identify the same reverse-shear budget, the full-Stokes residual threshold in Equation (43), the localized-obstacle bound in Equation (52), the plug-flow cancellation result in Equation (57), or the split trajectory diagnostic in Equation (62). Priority for such elementary identities is difficult to establish exhaustively, so these claims should be read as a cautious reformulation and extension of Parrenin–Hindmarsh theory, not as a new universal theorem about incompressible Stokes flow.

The strongest future novelty would come from applying the diagnostic to realistic Allan Hills geometry and history. A model that either proves a large positive order margin or demonstrates a trajectory crossing $\mathcal{J} = 0$ would resolve whether fixed roughness is dynamically adequate.

Coupling that calculation to observed layer attitudes, age reversals and fabric would connect a
 585 mathematical sign criterion to a specific paleoclimate archive.

9.5 Limitations

The analysis neglects diffusion of age markers, fracture and discontinuous displacement. It
 assumes a smooth two-dimensional velocity field and a single-valued accumulating inflow
 branch. The full-Stokes decomposition assumes isotropic scalar viscosity. It does not establish
 590 a universal mechanical bound on R_{FS} , because none follows from the Stokes equations alone.
 The localized-obstacle estimate depends on externally supplied bounds on minimum speed,
 deformation scale and reverse shear. The periodic cancellation result is first order and can be
 destroyed by asymmetry or nonlinear geometry.

Most importantly, the theorem diagnoses overturning of a continuous material isochrone.
 595 An observed age reversal can also result from faulting, unconformity, multiple source regions,
 freeze-on ice, sampling across a sharply dipping but discontinuous structure, or unresolved
 three-dimensional geometry. The order diagnostic should therefore be used with structural
 observations rather than as a stand-alone interpretation.

10 Conclusions

- 600 1. A steady two-dimensional isochrone preserves depositional order if and only if its signed
 order factor

$$\mathcal{J} = \frac{u}{u_{s0}} \left[1 + \dot{a}_0 u_{s0} \int_{x_0}^x \frac{u_z}{u^3} d\xi \right]$$

remains positive. The equivalent observable condition is $\tau_z < 0$, and in the ordered regime $\mathcal{J}$
 is the reciprocal of the established vertical thinning function.

- 605 2. Pointwise surface-fast shear, $u_z \geq 0$, is sufficient but not necessary. The exact condition is
 cumulative:

$$\mathcal{R} < 1 + \mathcal{P}.$$

Local extrusion is harmless unless its weighted path integral erases the unit depositional
 margin and all accumulated normal shear.

- 610 3. Full-Stokes mechanics does not guarantee the condition for arbitrary glacier geometry. Gravity-
 driven shear favors order, but longitudinal stress, nonhydrostatic pressure and horizontal
 gradients of vertical velocity can generate reverse shear. Stagnation and recirculation invalidate
 the one-way coordinate construction entirely.

- 615 4. The shallow-ice regime preserves order. A non-stagnant localized obstacle obeys the sufficient
 bound

$$\mathcal{R} \leq C \frac{r_s}{r^3} \frac{\dot{a}_0}{U} \frac{\ell_-}{d},$$

and small-amplitude periodic roughness about a uniform plug-flow background has no first-order net order change over a complete wavelength.

5. The theory converts the intuition that fixed obstacles produce draping rather than folds into a conditional, testable statement. In ordinary fast, shallow throughflow, boundary or rheological changes are effectively required to seed recumbent folding. In slow, rough, near-stagnant full-Stokes flow, fixed geometry remains a viable mechanism until the cumulative diagnostic is evaluated.

6. For COLDEX, the decisive next calculation is to integrate $\mathcal{P}$, $\mathcal{R}$ and $\mathcal{J}$ along reconstructed source-to-core trajectories at Allan Hills. This will distinguish steady topographic deformation from folding that requires ablation history, basal mass exchange, traction change, geometry change, anisotropy or three-dimensional flow.

Data availability

No new observational data are presented. The analytical results are fully specified by the equations in this manuscript. The numerical diagnostic requires a model velocity field and its vertical derivative.

Code availability

No software is included with this theoretical draft. Equations (60)–(62) provide a direct implementation specification for particle-tracking codes.

Author contribution

Bradley Paul Lipovsky developed the analysis and wrote the manuscript draft.

Competing interests

To be completed by the author before submission.

Acknowledgements

To be completed before submission. Support and contributions from NSF COLDEX and collaborating investigators should be documented here.

A Inverse-coordinate identities

Equation (8) can be written

$$\begin{pmatrix} \tau_x & \tau_z \\ \psi_x & \psi_z \end{pmatrix} \begin{pmatrix} dx \\ dz \end{pmatrix} = \begin{pmatrix} d\tau \\ d\psi \end{pmatrix}, \quad \det = 1. \quad (64)$$

645 The inverse transformation is

$$\begin{pmatrix} dx \\ dz \end{pmatrix} = \begin{pmatrix} \psi_z & -\tau_z \\ -\psi_x & \tau_x \end{pmatrix} \begin{pmatrix} d\tau \\ d\psi \end{pmatrix} = \begin{pmatrix} u & -\tau_z \\ w & \tau_x \end{pmatrix} \begin{pmatrix} d\tau \\ d\psi \end{pmatrix}. \quad (65)$$

Therefore

$$x_\tau|_\psi = u, \quad z_\tau|_\psi = w, \quad x_\psi|_\tau = -\tau_z, \quad z_\psi|_\tau = \tau_x. \quad (66)$$

The first two identities state that advancing one unit in age at fixed streamline follows the velocity for one unit of time. The latter two specify the equal-age tangent. Multiplication by $d\psi/dx_0 = \dot{a}_0$ yields Equation (19) and the corresponding vertical tangent component.

B Extension to steady variable-density flow

Below the firm, glacier ice is accurately treated as incompressible with nearly constant density. The coordinate construction extends to a steady variable-density flow satisfying

$$655 \quad \nabla \cdot (\rho \mathbf{u}) = 0. \quad (67)$$

Define the mass-flux streamfunction by

$$\rho u = \psi_z, \quad \rho w = -\psi_x. \quad (68)$$

Then

$$\frac{\partial(\tau, \psi)}{\partial(x, z)} = \rho(u\tau_x + w\tau_z) = \rho. \quad (69)$$

660 The mapping remains nonsingular for $\rho > 0$, and the same order interpretation follows after replacing volume flux by mass flux. The constant-density formula used in the main text is therefore not conceptually tied to strict incompressibility, although the weights in the cumulative integral acquire density factors.

C First-order periodic cancellation about plug flow

665 Let

$$u(x, z; \epsilon_b) = U_b + \epsilon_b u_1(x, z) + O(\epsilon_b^2), \quad U_b > 0, \quad (70)$$

where U_b is spatially uniform and u_1 is periodic in x with wavelength $2\pi/k$ and zero horizontal mean at each elevation. Then

$$u_z = \epsilon_b u_{1,z} + O(\epsilon_b^2), \quad u^{-3} = U_b^{-3} + O(\epsilon_b). \quad (71)$$

Because the background integrand is identically zero, the $O(\epsilon_b)$ displacement of the trajectory affects the integral only at $O(\epsilon_b^2)$. Evaluating the first-order term along the unperturbed streamline $z = z_0$ gives

$$\int_x^{x+2\pi/k} \frac{u_z}{u^3} d\xi = \frac{\epsilon_b}{U_b^3} \int_x^{x+2\pi/k} u_{1,z}(\xi, z_0) d\xi + O(\epsilon_b^2) = O(\epsilon_b^2), \quad (72)$$

because differentiation with respect to z preserves the zero horizontal mean of u_1 . This establishes Equation (57). For a vertically sheared background, trajectory displacement can act on the nonzero background integrand at first order, so the cancellation does not follow without additional assumptions.

References

- Alley, R. B., A. J. Gow, D. A. Meese, J. J. Fitzpatrick, E. D. Waddington, and J. F. Bolzan (1997). Grain-scale processes, folding, and stratigraphic disturbance in the GISP2 ice core. In: *Journal of Geophysical Research: Oceans* 102.C12, pp. 26819–26830. DOI: 10.1029/96JC03836.
- Bintanja, R. (1999). On the glaciological, meteorological, and climatological significance of Antarctic blue ice areas. In: *Reviews of Geophysics* 37.3, pp. 337–359. DOI: 10.1029/1999RG900007.
- Blatter, H. (1995). Velocity and stress fields in grounded glaciers: a simple algorithm for including deviatoric stress gradients. In: *Journal of Glaciology* 41.138, pp. 333–344. DOI: 10.3189/S002214300001621X.
- Bons, P. D., Y. Hu, M.-G. Llorens, S. Franke, N. Stoll, I. Weikusat, J. Westhoff, and Y. Zhang (2025). Folding due to anisotropy in ice, from drill-core-scale cloudy bands to km-scale internal reflection horizons. In: *The Cryosphere* 19, pp. 5095–5109. DOI: 10.5194/tc-19-5095-2025.
- Bons, P. D., D. Jansen, F. Mundel, C. C. Bauer, T. Binder, O. Eisen, M. W. Jessell, M.-G. Llorens, F. Steinbach, D. Steinhage, and I. Weikusat (2016). Converging flow and anisotropy cause large-scale folding in Greenland’s ice sheet. In: *Nature Communications* 7, p. 11427. DOI: 10.1038/ncomms11427.
- Cuffey, K. M. and W. S. B. Paterson (2010). *The Physics of Glaciers*. 4th ed. Amsterdam: Elsevier.
- Gudmundsson, G. H. (1997a). Basal-flow characteristics of a linear medium sliding frictionless over small bedrock undulations. In: *Journal of Glaciology* 43.143, pp. 71–79. DOI: 10.3189/S0022143000002823.
- (1997b). Basal-flow characteristics of a non-linear flow sliding frictionless over strongly undulating bedrock. In: *Journal of Glaciology* 43.143, pp. 80–89. DOI: 10.3189/S0022143000002835.

- Higgins, J. A., A. V. Kurbatov, N. E. Spaulding, E. J. Brook, D. S. Introne, L. M. Chimiak, Y. Yan, P. A. Mayewski, and M. L. Bender (2015). Atmospheric composition 1 million years ago from blue ice in the Allan Hills, Antarctica. In: *Proceedings of the National Academy of Sciences* 112.22, pp. 6887–6891. DOI: 10.1073/pnas.1420232112.
- Hindmarsh, R. C. A., G. J.-M. C. Leysinger Vieli, and F. Parrenin (2009). A large-scale numerical model for computing isochrone geometry. In: *Annals of Glaciology* 50.51, pp. 130–140. DOI: 10.3189/172756409789097450.
- Hindmarsh, R. C. A., G. J.-M. C. Leysinger Vieli, M. J. Raymond, and G. H. Gudmundsson (2006). Draping or overriding: The effect of horizontal stress gradients on internal layer architecture in ice sheets. In: *Journal of Geophysical Research: Earth Surface* 111.F2, F02018. DOI: 10.1029/2005JF000309.
- Hudleston, P. J. (2015). Structures and fabrics in glacial ice: A review. In: *Journal of Structural Geology* 81, pp. 1–27. DOI: 10.1016/j.jsg.2015.09.003.
- Jacobson, H. P. and E. D. Waddington (2004). Recumbent folding in ice sheets: a core-referential study. In: *Journal of Glaciology* 50.168, pp. 3–16. DOI: 10.3189/172756504781830204.
- (2005). Recumbent folding of divide arches in response to unsteady ice-divide migration. In: *Journal of Glaciology* 51.173, pp. 201–209. DOI: 10.3189/172756505781829412.
- Jennings, S. J. A. and M. J. Hambrey (2021). Structures and deformation in glaciers and ice sheets. In: *Reviews of Geophysics* 59.3, e2021RG000743. DOI: 10.1029/2021RG000743.
- Kehrl, L., H. Conway, N. Holschuh, S. Campbell, A. V. Kurbatov, and N. E. Spaulding (2018). Evaluating the duration and continuity of potential climate records from the Allan Hills Blue Ice Area, East Antarctica. In: *Geophysical Research Letters* 45.9, pp. 4096–4104. DOI: 10.1029/2018GL077511.
- Kirkpatrick, L. R., A. J. Carter, J. Marks-Peterson, S. Shackleton, and T. J. Fudge (2025). Three-dimensional multitrack electrical conductivity method for interpretation of complex ice core stratigraphy. In: *Journal of Glaciology* 71, e105. DOI: 10.1017/jog.2025.10081.
- Leysinger Vieli, G. J.-M. C., C. Martín, R. C. A. Hindmarsh, and M. P. Lüthi (2018). Basal freeze-on generates complex ice-sheet stratigraphy. In: *Nature Communications* 9, p. 4669. DOI: 10.1038/s41467-018-07083-3.
- Martín, C., G. H. Gudmundsson, H. D. Pritchard, and O. Gagliardini (2009). On the effects of anisotropic rheology on ice flow, internal structure, and the age–depth relationship at ice divides. In: *Journal of Geophysical Research: Earth Surface* 114.F4, F04001. DOI: 10.1029/2008JF001204.
- Martín, C., R. C. A. Hindmarsh, and F. J. Navarro (2009). On the effects of divide migration, along-ridge flow, and basal sliding on isochrones near an ice divide. In: *Journal of Geophysical Research: Earth Surface* 114.F2, F02006. DOI: 10.1029/2008JF001025.
- Moffatt, H. K. (1964). Viscous and resistive eddies near a sharp corner. In: *Journal of Fluid Mechanics* 18.1, pp. 1–18. DOI: 10.1017/S0022112064000015.
- Parrenin, F. (2013). “Sur l’âge de la glace et des bulles d’air dans les calottes polaires”. Thesis. Grenoble, France: Université Joseph Fourier.
- Parrenin, F., A. Chung, and C. Martín (2025). age_flow_line-1.0: a fast and accurate numerical age model for a pseudo-steady flow tube of an ice sheet. In: *Geoscientific Model Development* 18, pp. 8203–8216. DOI: 10.5194/gmd-18-8203-2025.

Parrenin, F. and R. C. A. Hindmarsh (2007). Influence of a non-uniform velocity field on isochrone geometry along a steady flowline of an ice sheet. In: *Journal of Glaciology* 53.183, pp. 612–622. doi: 10.3189/002214307784409298.

745 Parrenin, F., R. C. A. Hindmarsh, and F. Rémy (2006). Analytical solutions for the effect of topography, accumulation rate and lateral flow divergence on isochrone layer geometry. In: *Journal of Glaciology* 52.177, pp. 191–202. doi: 10.3189/172756506781828728.

Schoof, C. and R. C. A. Hindmarsh (2010). Thin-film flows with wall slip: An asymptotic analysis of higher order glacier flow models. In: *The Quarterly Journal of Mechanics and Applied Mathematics* 63.1, pp. 73–114. doi: 10.1093/qjmam/hbp025.

750 Shackleton, S., V. Hishamunda, L. Davidge, E. J. Brook, J. Marks-Peterson, A. Carter, S. Aarons, A. V. Kurbatov, D. S. Introne, Y. Yan, I. M. Nesbitt, C. Buizert, E. J. Steig, A. J. Schauer, J. Morgan, P. D. Neff, J. A. Epifanio, J. P. Severinghaus, M. L. Bender, and J. A. Higgins (2025). Miocene and Pliocene ice and air from the Allan Hills blue ice area, East Antarctica. In: *Proceedings of the National Academy of Sciences* 122.44, e2502681122. doi: 10.1073/pnas.2502681122.

755 Spaulding, N. E., V. B. Spikes, G. S. Hamilton, P. A. Mayewski, N. W. Dunbar, R. P. Harvey, J. Schutt, and A. V. Kurbatov (2012). Ice motion and mass balance at the Allan Hills blue-ice area, Antarctica, with implications for paleoclimate reconstructions. In: *Journal of Glaciology* 58.208, pp. 399–406. doi: 10.3189/2012JoG11J176.

760 Stoll, N., M. Shaya, L. Kirkpatrick, J. Freitag, V. Hishamunda, J.-M. Manos, D. Jansen, I. Weikusat, J. Fegyveresi, B. Lipovsky, E. J. Brook, S. Shackleton, J. A. Higgins, and T. J. Fudge (2026). Fabric, texture, and bubble characteristics of the million-year old Allan Hills blue ice core ALHIC1901. In: *Journal of Geophysical Research: Solid Earth* 131.5, e2026JB033890. doi: 10.1029/2026JB033890.

765 Thorsteinsson, T. and E. D. Waddington (2002). Folding in strongly anisotropic layers near ice-sheet centers. In: *Annals of Glaciology* 35, pp. 480–486. doi: 10.3189/172756402781816708.

Waddington, E. D. (2010). Life, death and afterlife of the extrusion flow theory. In: *Journal of Glaciology* 56.200, pp. 973–996. doi: 10.3189/002214311796406022.

Waddington, E. D., J. F. Bolzan, and R. B. Alley (2001). Potential for stratigraphic folding near ice-sheet centers. In: *Journal of Glaciology* 47.159, pp. 639–648. doi: 10.3189/172756501781831756.

770 Wolovick, M. J. and T. T. Creyts (2016). Overturned folds in ice sheets: Insights from a kinematic model of traveling sticky patches and comparisons with observations. In: *Journal of Geophysical Research: Earth Surface* 121.5, pp. 1065–1083. doi: 10.1002/2015JF003698.

775 Wolovick, M. J., T. T. Creyts, W. R. Buck, and R. E. Bell (2014). Traveling slippery patches produce thickness-scale folds in ice sheets. In: *Geophysical Research Letters* 41.24, pp. 8895–8901. doi: 10.1002/2014GL062248.

Yan, Y., M. L. Bender, E. J. Brook, H. M. Clifford, P. C. Kemeny, A. V. Kurbatov, S. Mackay, P. A. Mayewski, J. Ng, J. P. Severinghaus, and J. A. Higgins (2019). Two-million-year-old snapshots of atmospheric gases from Antarctic ice. In: *Nature* 574, pp. 663–666. doi: 10.1038/s41586-019-1692-3.

780 Zhang, Y., T. Sachau, S. Franke, H. Yang, D. Li, I. Weikusat, and P. D. Bons (2024). Formation mechanisms of large-scale folding in Greenland's ice sheet. In: *Geophysical Research Letters* 51.16, e2024GL109492. DOI: 10.1029/2024GL109492.